School of Mechanical Engineering	UNIVERSITY OF LEEDS

MECH5845M Professional Project

Scoping and Planning Document

Student Name: **Nisarg Mahendrabhai Makwana**	
Programme of Study: **MSc Advanced Mechanical Engineering**	
Provisional Title of Project: **Flow analysis of a fluid mixer in liquid flow cell for TEM(Transmission Electron Microscope) study**	
Name of External Company (if any)**:** **Hummingbird Scientific**	
Supervisor(s) Name(s): Prof. **Nikil Kapur**	
Type of Project: **Designing and Analysis of Computational Model**	
NOTES TO STUDENT: *a) The font used should be no smaller than Arial 11pt or equivalent, with 1½ line spacing and margins no smaller than 25mm (top, right and bottom) and 38 mm (left)* *b) Ensure you have discussed the content with your supervisor well in advance of the deadline for submission.* *c) An **electronic version** of this report in pdf format must be submitted via the module folder in Minerva; with the filename of the format: SURNAME_FirstName_SID.pdf (e.g. SMITH_Paul_234567891.pdf).*	
Signature of Student: Nisarg	**Date: 14/02/2022**

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Contents

1. Scope of the project 1

1.1 Project Aim 1

1.2 Project Objectives 1

1.3 Project Deliverables 2

2. “Lite” Preliminary Literature Review 2

2.1 Flow Cell: 2

2.2 Flow Properties in Flow Cell and Study: 3

2.3 Numerical Study: 3

2.4 Mixing Methodologies: 4

2.5 Summary: 5

3. Project Work Plan and Ethical Considerations 5

3.1 Tasks, Milestones and Timeline 5

3.2 Project Risk assessment 7

4 References 7

Appendix A. Ethical Review Guidance 9

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1. Scope of the project

Microfluidic devices have been widely used for particle synthesis over the last few years, primarily for inorganic, polymeric, and metallic particles of various materials and functionalities[11]. These devices have had a considerable impact on the fields of biomedical diagnostics and drug development and are extensively applied in the food and chemical industries[6]. The small scale of the flow channels in such devices increases the surface-to-volume ratio and thus is advantageous for multiple applications.

For my project, I will be working on the mixing methodologies part of the Liquid flow cell design to obtain a better mixing index, Which is a crucial factor in obtaining a clear and smooth viewing interface for image spectroscopy study in TEM(Transmission Electron Microscope). Also, there is a limited amount of data available of the flow pattern in microfluidic systems, which will be achieved in this project.

1.1 Project Aim

The aim of the current project is to obtain thorough and rapid mixing of continuous microfluidic flow in flow cells applicable for TEM(Transmission Electron Microscope) studies for image spectroscopy.

1.2 Project Objectives

Some of the main objectives of this Project are listed as follows.

1. Literature Review and Background Research of the Flow Cell and Flow Mixers for Microfluidic flow systems.
2. Defining the requirements, properties and functions of the flow cell device currently in use and discussing it with project supervisor and associated people working in this field.
3. Learning Analysis Software like COMSOL Multiphysics (Transport of Diluted Specimen and Laminar Flow Models)
4. Building a computational model of the CAD Geometry of the flow cell using a CAD software (Solidworks) and analysis study in COMSOL Multiphysics 5.5.
5. Developing a mixer design to add external turbulence in the low reynolds number flow to improve the flow mixing index as the mixing occuring in here is mostly from diffusion which is an inherently slow process[6].
6. Running a parametric study and result visulaization and looking for the effect caused by flow rate on mixing.

7. Testing of modified computational model using the new geometry on the current flow cell to analyse the performance and suggest the new design changes which will improve the device performance and obtain better results.
8. Deliver a oral presentation of the project on the MSc conference week.
9. Report to be drafted, proofread and submitted with the result, achievements, and meeting logs by the last week of july.

1.3 Project Deliverables

1. A potential CAD design for the flow cell mixing component used for TEM image sampling.
2. A computational model for the microfluidic study
3. Obtain new data regarding the effect of flow rate on mixing index for microfluidic flow system.
4. Oral presentation of the project
5. A written report

2. “Lite” Preliminary Literature Review

The use of microfluidic flow systems to control samples of a micrometer size in liquids has exponentially increased in various research disciplines such as analytical chemistry, biology, and medicine [1, 2]. Mixing in microfluidic channels has great potential because it can control the number of droplets, the concentration of chemicals, and the sorting of liquids [3]. Recently, microfluidic mixtures have been used in drug development and medical diagnostics [4, 5].

2.1 Flow Cell:

A flow cell is a device that is designed in such a way that enhances and facilitates the mixing of continuous microscopic fluid flow samples in a specified path of flow in spectroscopic studies on TEM(Transmission Electron Microscope)[9]. In these devices, new samples are constantly replaced to ensure that there is no interruption, distortion, or damage occurring with the signal.

Figure 1 Hummingbird Scientific's TEM liquid flow holder.[9]

A flow cell designed by Hummingbird Scientist as a liquid TEM liquid cell(Figure 1) has the function of keeping the specimen in a liquid environment while the chamber is pumped down to quite high vaccum[9]. This device also provides researchers precise control over the microfluidic flow systems over the surface of the specimen[9].

However, the Flow cell is highly useful in correlative studies in X-Ray Studies and Optical Microscopy and SEM Spectroscopy(Figure 2)[10].

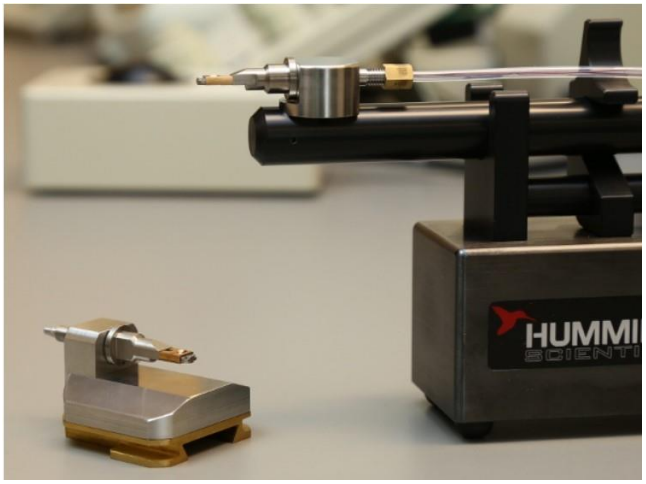

Figure 2 Hummingbird Scientific's SEM liquid cell[10].

2.2 Flow Properties in Flow Cell and Study:

The flow occurring in a flow cell is on a microscopic scale; thus, the flow occurring has a low Reynolds number, and for such kind of flow, no turbulence is to be discovered. In such a case, microscale mixing is necessary[11]. As for macroscopic mixing, a turbulent component in the flow is a requirement. It is physically impossible to generate turbulent flow for such a low Reynolds number, and mixing In laminar flow needs to be done. In such laminar flows, the advection(movement in the direction of flow) component is extremely low, and most of the mixing of the two liquid flows occurs due to diffusion(due to concentration gradients within the fluid)[11].

One such flow condition is assumed by one research study where a steady-state flow has a constant flow velocity of fluids and different solute concentrations that are mixing in a T-shaped channel, and appropriate mixing occurs at the boundary of the two fluids[7]. Here, to determine the status of the flow, the Reynolds number($Re = \rho Ud/\mu$, where U is the mean velocity, d is the width of a cell, and μ and ρ are the viscosity and fluid density), can be used. To reach the appropriate concentration of the influent after mixing, a precise flow pattern is required as it determines the flow rate and flow volume[11]. In some regions, eddies are developed where the channel width is high, and these patterns inside the flow cells determine how fast the concentration of analyte fluid reaches the influent(injected fluid)[11]. The flow pattern changes with the change in flow geometry and flows velocity, and this change needs to be studied for different mixer geometry and flow rates[11].

2.3 Numerical Study:

Numerically, a time Dependent advection-diffusion equation is studied to obtain the flow response for an incompressible flow. This causes a simulation of the transport of dilute species through a solvent[11]. Advection Diffusion Equation:-

$$\frac{\partial c}{\partial t} = D\nabla^2 c - v \cdot \nabla c$$

A transport because of diffusion is controlled via a concentration gradient at one particular place. The underlying flow fields are responsible for the transport due to advection, and this is determined through a solution of steady-state Navier-Stokes Equation, with appropriate boundary conditions for laminar fluid flow[11]. Navier-Stokes Equation:-

$$\rho v \cdot \nabla v = -\nabla p + \mu \nabla^2 v \qquad \& \qquad \nabla \cdot v = 0$$

2.4 Mixing Methodologies:

Many studies present schemes by providing flow samples through Multi-channels, holes, and channels with obstacles in microfluidic devices. The main two Methodologies are discussed as follows:

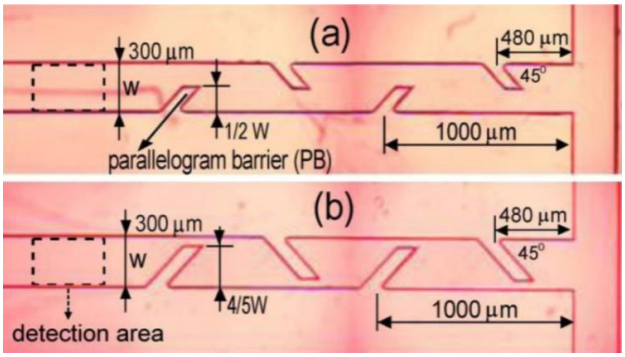

Figure 3 Microscopic image of microfludic mixer with parallelogram barriers in mixing channel[13].

Active Mixing Methods: In active mixing schemes, the flow performance is enhanced by the application of external force to develop and assist in the diffusion of flow[12]. Generally, this scheme is implemented by the incorporation of mechanical actuation and generally has a moving component. Acoustic, dielectrophoretic, magneto-hydrodynamic flow, electrokinetic instability(Figure 3) are the most efficient active mixing methodologies currently in use.

Passive Mixing Methods: In passive mixing schemes, an approach of increasing the contact area of the mixing fluids so that fluids interact at multiple points, Also the mixing performance can be improved by increasing the contact time of liquids[6]. This scheme doesn't have any moving component and requires no energy other than a pressure head to drive the flow. Here the mixing relies on chaotic interaction by an increase of contact are and time[6]. Some of the most efficient passive mixing methods are zigzag channels(Figure 4), an unbalanced driving force in serpentine structure, and T or Y Mixing channels.

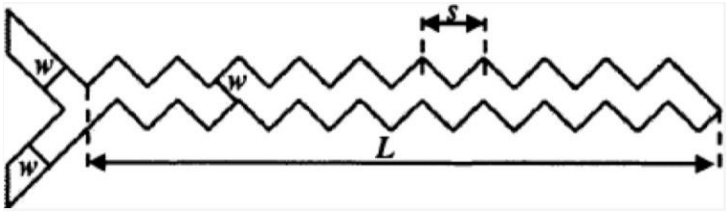

Figure 4 Dimensions of the microfluidic system integrating a "Y" junction, with w, the width of the channel; s, the linear length of the periodic step; and L, the linear length of the zigzag microchannel.[14]

2.5 Summary:

In current studies regarding microfluidic flow systems, data of flow interaction, mixing, and flow behavior is still limited. As discussed in Section 2.2, In microfluidic devices like flow cells, the flow pattern changes with the change in flow geometry and flow velocity while generating maximum mixing performance is a challenge seen. And this data for different mixing conditions new designs at different flow rates will be studied in this project with parameters like mixing time, device volume, and influent concentration. Further, the data obtained will be verified with the Actual base design that is currently in use.

3. Project Work Plan and Ethical Considerations

3.1 Tasks, Milestones and Timeline

3.1.1 Milestones

1. Development of computational model design of the mixer design
2. Computational analysis of the flow model on COMSOL multiphysics
3. Generated results for the preliminary design
4. Modified and optimized the design of the model and generated required results.
5. Compared results of the model with actual device in use
6. Completion of the project and final report

3.1.2 Tasks & Timeline

1. Commence working on the background research and literature review.
2. Submitting the final draft of scoping documents.
3. Start discussion and study on design specifications and mixing conditions for the flow cell.
4. Begin working on CAD Model with base design for further analysis.
5. Modify the device computationally with correct boundary conductions.
6. Start working on COMSOL multiphysics software.
7. Begin model study, flow analysis, and generate appropriate results
8. Completing the liquid flow cell design, with updated & modified design, and generating appropriate results.
9. Comparing the experimental data with current computational design data
10. Writing a final report of the project.

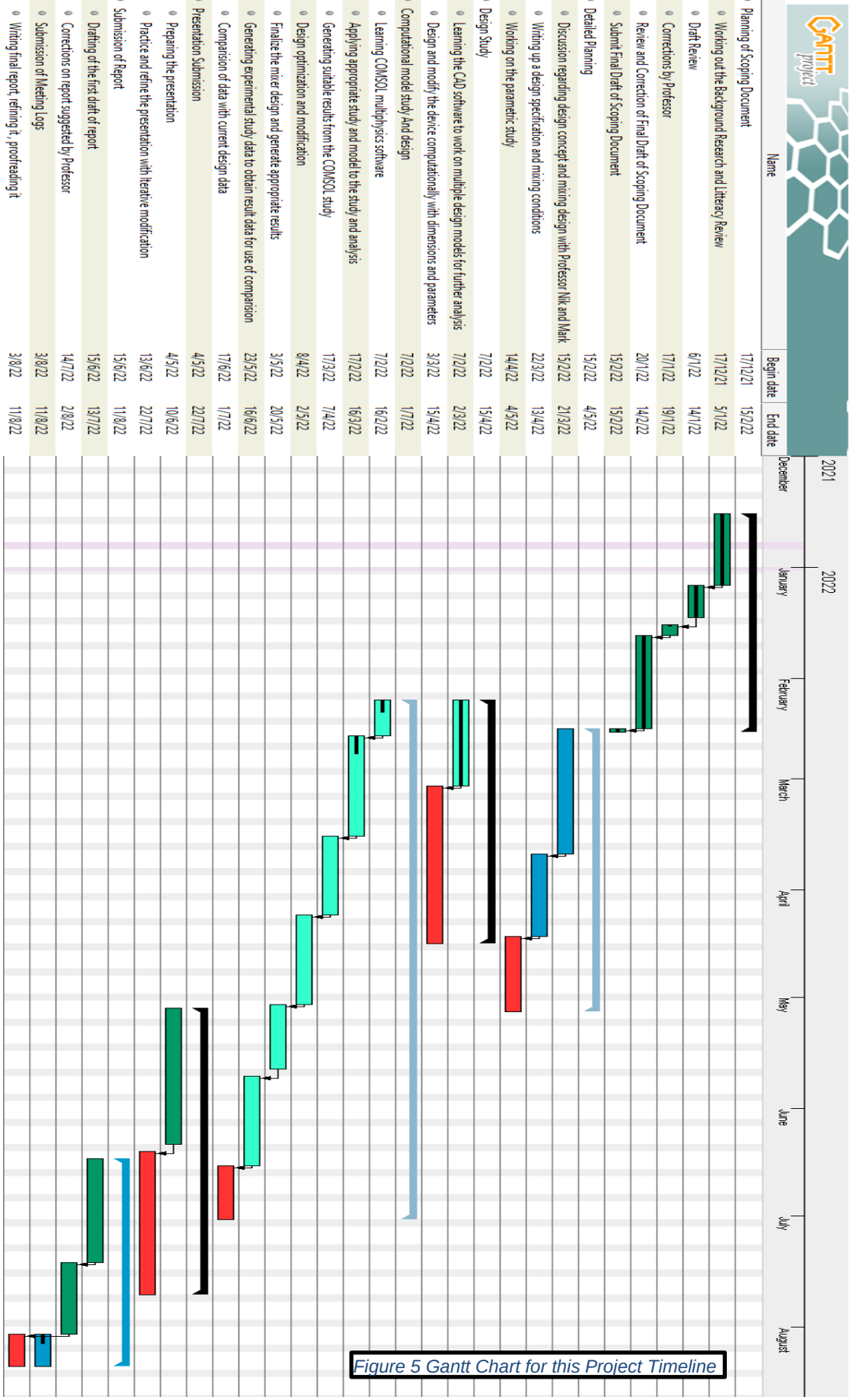

Figure 5 Gantt Chart for this Project Timeline

3.2 Project Risk assessment

The Following Risks and Hindrances that may occur which can prevent or delay the project outcomes are as follows:

Risk	Likelihood	Severity	Mitigating Circumstance
Health Issue	Likely	Moderate	Visiting Doctor and Taking Health precautions
Generated data loss or corruption	Likely	Extreme	Backing up data on drive and external hardware device.
Final report data corruption	Less likely	Moderate	Backing up data on drive and external hardware device.
Computer gets Stolen	Less Likely	High	Backing up data on drive and external hardware device.
Crashing of Software	High	High	Working on a Computer clusters and working with parameters where low computational power is required.
Incorrect Results	High	Moderate	Regular meeting and discussion with guide and continious verification of data.
Software study more complicated than assumed	High	Moderate	Allocating more buffer time to the model study and application.
Geometry optimization takes long	Very High	Fair	Continuous design modification and fast iteration will be required to be done.
Availability of LAB and Meetings with project associated people	Likely	Less	Proper meeting pre-planning and maintaining proper data and records of each session.
Generated results differ from expected results	Less Likely	Less	Good literature review and write an appropriate reason for the cause.

For this project, Ethical Approval is not required as there is no involvement of human participation and data that can be acquired from them. Also, there is no significant environmental impact that may occur from this project, so the University of Leeds ethical approval is not required.

4 References

A list of all the references used in this report are listed as follows:

1. N.-T. Nguyen and Z. Wu, "Micromixers—a review," Journal of Micromechanics and Microengineering 15, R1 (2004).
2. D. Mark, S. Haeberle, G. Roth, F. von Stetten, and R. Zengerle, "Microfluidic lab-on-a-chip platforms: requirements, characteristics and applications," Chemical Society Reviews 39, 1153-1182 (2010).
3. Y.-C. Tan, J. S. Fisher, A. I. Lee, V. Cristini, and A. P. Lee, "Design of microfluidic channel geometries for the control of droplet volume, chemical concentration, and sorting," Lab on a Chip 4, 292-298 (2004).
4. I. Glasgow and N. Aubry, "Enhancement of microfluidic mixing using time pulsing," Lab on a Chip 3, 114-120 (2003).
5. K. S. Elvira, X. C. i Solvas, and R. C. Wootton, "The past, present and potential for microfluidic reactor technology in chemical synthesis," Nature Chemistry 5, 905-915 (2013).
6. Lee, C.-Y., Chang, C.-L., Wang, Y.-N. and Fu, L.-M. 2011. Microfluidic Mixing: A Review. International Journal of Molecular Sciences. 12(5), pp.3263–3287.
7. Park, G., Han, D., Kim, G., Shin, S., Kim, K., Park, J. and Park, Y. 2017. Visualization and label-free quantification of microfluidic mixing using quantitative phase imaging. Applied Optics. 56(22),p.6341.
8. Charmet, J., Arosio, P. and Knowles, T. 2018. Microfluidics for Protein Biophysics. *Journal of Molecular Biology*. 430(5),pp.565-580.
9. Anon 2022. Overview | Hummingbird Scientific. *Hummingbird Scientific*. [Online]. [Accessed 15 February 2022]. Available from: https://hummingbirdscientific.com/products/liquid/#.
10. Anon 2022. Overview | Hummingbird Scientific. *Hummingbird Scientific*. [Online]. [Accessed 15 February 2022]. Available from: https://hummingbirdscientific.com/sem-liquid-overview/.
11. Pike, D., Kapur, N., Millner, P. and Stewart, D. 2012. Flow Cell Design for Effective Biosensing. *Sensors*. 13(1),pp.58-70.
12. Meijer, H., Singh, M., Kang, T., den Toonder, J. and Anderson, P. 2009. Passive and Active Mixing in Microfluidic Devices. *Macromolecular Symposia*. 279(1),pp.201-209.
13. Tai, C., Yang, R., Huang, M., Liu, C., Tsai, C. and Fu, L. 2006. Micromixer utilizing electrokinetic instability-induced shedding effect. *ELECTROPHORESIS*. 27(24),pp.4982-4990.

Appendix A. Ethical Review Guidance

This is a University requirement. For guidance please see Figure A1 and discuss it with your supervisor. You do not need to include this appendix – only the ethics part of the project risk assessment is required.

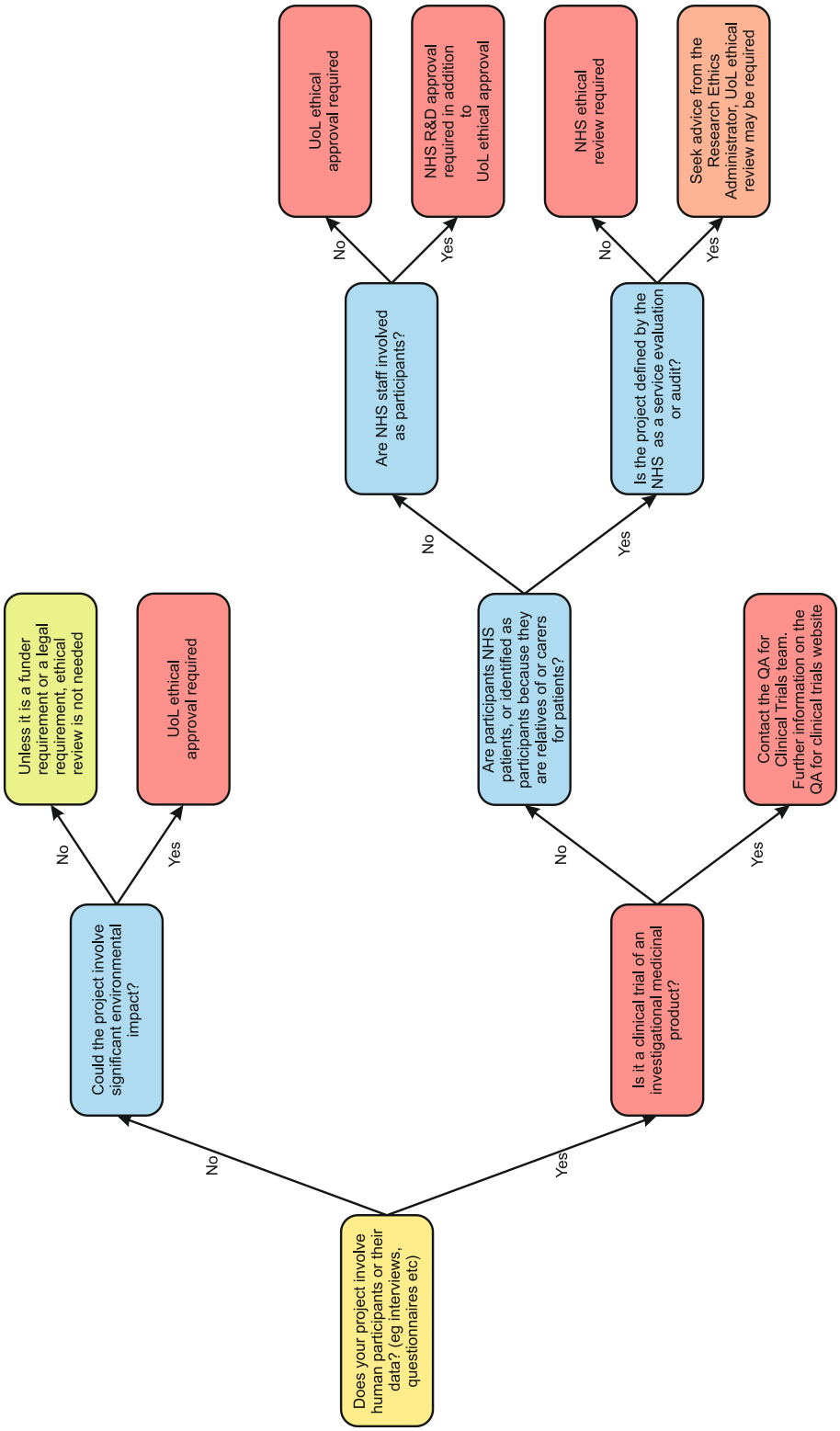

Figure A1: Research Ethics Flow Chart